

Software Test Plan (STP)

Project: Collaborative Study Material & Quiz Platform (CSQP)

Authors: Harshith R Reddy (PES2UG24CS188), Kritika Mahapatra (PES2UG24CS239), Krishna Manoj (PES2UG24CS236), John S Thomas (PES2UG24CS207)

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Team Member Contribution

No.	Team Member	Assigned Work
1	Harshith R Reddy (PES2UG24CS188)	Introduction, test items scoping, test environment, and external interfaces
2	Kritika Mahapatra (PES2UG24CS239)	Overall test strategy, test levels, criteria, deliverables, and team responsibilities
3	Krishna Manoj (PES2UG24CS236)	Functional test case design (material sharing, question bank, quiz & grading)
4	John S Thomas (PES2UG24CS207)	Security validation (5.1), non-functional performance testing, and RTM mapping

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1. Introduction

1.1 Purpose: This Software Test Plan (STP) defines the testing objectives, scope, methodology, environment, schedule, and test cases for the Collaborative Study Material & Quiz Platform (CSQP). It ensures that all functional features and quality benchmarks defined in the SRS v1.0 and SADS v1.0 are verified systematically before deployment.

1.2 Scope: Testing covers the React web frontend, the Node/Express REST API backend, MongoDB data handling, and Python analytical routines. Features tested include college email registration, JWT auth, RBAC permissions, notes upload and versioning, question curation with instructor approval, anti-cheat timed quizzes, and automated scoring. Testing excludes vendor hardware maintenance and third-party mail server internals.

1.3 References: CSQP SRS v1.0, CSQP Software Architecture & Design Specification (SADS) v1.0, IEEE 829 Standard for Software Test Documentation, and WCAG 2.1 AA Accessibility Guidelines.

1.4 Definitions and Acronyms: CSQP (Collaborative Study Material & Quiz Platform), STP (Software Test Plan), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), RBAC (Role-Based Access Control), JWT (JSON Web Token), UAT (User Acceptance Testing).

2. Test Items

The software items targeted for verification correspond to our core architectural components:

- **AuthModule:** Email registration, verification token expiration, login validation, JWT issuance, and account logout.
- **MaterialModule:** Upload validation (type and 25MB limit), topic filtering, star rating computation, and version history retention.
- **QuestionBankModule:** Creation of MCQ, True/False, and Short-Answer questions, CSV import parsing, and instructor review queue.
- **QuizModule & AutoGrading:** Quiz generation (curated and sampled), countdown timers, clipboard disabling, and immediate scoring (< 2s).
- **AnalyticsModule:** Score trend computations, item difficulty analysis, opt-in leaderboard filters, and PDF/CSV report exports.

3. Features to be Tested

Features mapped to SRS requirement IDs:

- **Authentication & Security:** CSQP-F-001 (Registration/Verification), CSQP-F-002 (JWT Login), CSQP-F-003 (RBAC Enforcement), CSQP-F-004 (Password Reset), CSQP-SR-005 (Account Lockout).
- **Study Materials:** CSQP-F-010 (File Upload & Constraints), CSQP-F-011 (Search & Filters), CSQP-F-012 (Ratings & Comments), CSQP-F-013 (Version Retention).
- **Question Bank & Quizzes:** CSQP-F-020 (Question Contribution), CSQP-F-021 (Question Types), CSQP-F-022 (Bulk Import), CSQP-F-023 (Instructor Approval), CSQP-F-024 (Quiz Assembly).
- **Quiz Attempts & Grading:** CSQP-F-030 (Countdown & Auto-Submit), CSQP-F-031 (Objective Auto-Grading), CSQP-F-032 (Randomization & Clipboard Lock), CSQP-F-033 (Single Active Attempt).
- **Analytics & Non-Functional:** CSQP-F-040 (Student Dashboard), CSQP-F-041 (Instructor Analytics), CSQP-F-042 (Leaderboard), CSQP-F-043 (Export Reports), CSQP-NF-001 ($p90 \leq 2s$ Latency), CSQP-NF-004 (WCAG 2.1 AA).

4. Features Not to be Tested

- Third-party SMTP email delivery network latency and ISP spam filtering mechanisms.
- Internal MongoDB storage engine clustering and low-level disk I/O routines.
- External browser rendering engine source code and hardware-level driver failures.
- Institutional Student Information Systems (SIS) integrations outside basic profile fields.

5. Test Approach / Strategy

Testing is organized across distinct verification levels and quality categories:

- **Unit Testing:** Focuses on isolated controller methods, data validation utilities, password hashing functions, and objective grading algorithms using Jest and Mocha.
- **Integration Testing:** Validates communication between Express API routes, Mongoose database schemas, and inter-service RPC calls to the Python analytics worker.
- **System Testing:** Executes end-to-end user workflows, such as question submission by a student, moderation by an instructor, quiz generation, timed attempt execution, and dashboard updating.
- **User Acceptance Testing (UAT):** Verification conducted with representative student and instructor users against the acceptance criteria defined in the SRS.

Entry and Exit Criteria: Testing begins once a stable build is deployed to the local staging environment with mock seed data.

Testing concludes when 100% of planned test cases are executed, all High-priority functional tests pass, and zero critical defects remain open.

5.1 Security Validation

To satisfy security objectives SEC-OBJ-01 through 03, dedicated security tests are enforced:

- **Password Security:** Verify that passwords are never logged or stored in plain text, and that bcrypt hashes include unique salt rounds (CSQP-SR-002).
- **Token Expiration & Refresh:** Validate that access tokens expire after 1 hour and that refresh tokens rotate upon renewal (CSQP-SR-003).
- **API Role Guards:** Probe restricted endpoints without proper role headers to verify that unauthorized requests receive HTTP 403 Forbidden (CSQP-SR-004).
- **Brute-Force Lockout:** Submit 5 consecutive invalid passwords for a user account to confirm that the 6th attempt is blocked for 15 minutes and logged in audit logs (CSQP-SR-005).
- **Quiz Integrity Checks:** Confirm that answer keys are omitted from client payloads during active attempts, questions/options are shuffled, and copy-paste clipboard events are prevented (CSQP-F-032, 033).

6. Test Environment

- **Client Devices:** Desktop and laptop browsers (Chrome, Firefox, Safari) and mobile responsive viewports.
- **Backend Runtime:** Node.js (v18+) with Express server and Python 3.10+ analytical service.
- **Database:** Local MongoDB instance (v6+) seeded with test accounts, questions, and attempt records.
- **Testing Tools:** Postman / Supertest for API integration testing, Jest for unit tests, k6 / Apache JMeter for performance tests, and Chrome Lighthouse for accessibility.

7. Test Schedule & Milestones

Milestone Phase	Start Date	End Date	Deliverable / Output
Test Plan & Test Case Design	15-09-2026	18-09-2026	Master STP document and test case specifications
Environment Setup & Mock Data	19-09-2026	20-09-2026	Seeded MongoDB database and test fixtures
Unit & Integration Testing	21-09-2026	24-09-2026	Automated API test execution and bug logging
System & Security Testing	25-09-2026	26-09-2026	RBAC, anti-cheating, and load test validation
User Acceptance & Final Report	27-09-2026	28-09-2026	UAT sign-off and Final Test Summary Report

8. Test Deliverables

Deliverables include this Test Plan (STP), detailed test cases, automated test scripts (Jest/Supertest/Postman), defect triage logs in GitHub Issues, and the Final Test Summary Report.

9. Roles and Responsibilities

- **Harshith R Reddy (QA Lead):** Oversees test execution, coordinates environment setup, and reviews test completion criteria.
- **Kritika Mahapatra (Test Engineer):** Executes UI responsive tests, study material sharing flows, and question bank workflows.
- **Krishna Manoj (Test Engineer):** Executes quiz attempt timers, auto-grading calculations, and manual review queues.
- **John S Thomas (Security & Performance QA):** Performs JWT/RBAC security scans, account lockout tests, and latency benchmarking.

10. Risks and Mitigation

- *Risk:* Delay in feature readiness. *Mitigation:* Run automated smoke tests early on authentication and question bank modules.
- *Risk:* Test data pollution during concurrent tests. *Mitigation:* Implement teardown scripts that re-seed MongoDB before every suite.
- *Risk:* Python service response lag. *Mitigation:* Mock background analytics endpoints during primary API functional testing.

11. Assumptions & Dependencies

We assume an operational SMTP gateway or local mock email service for OTP verification, a seeded pool of at least 50 questions, and network access to local API servers.

12. Suspension & Resumption Criteria

Testing is suspended if the authentication service is completely unavailable or if a blocking defect prevents more than 30% of planned test cases from running. Testing resumes once the blocking bug is resolved and verified.

13. Test Case Management & Traceability

Every test case is traceable to a specific SRS functional or non-functional requirement via the Requirements Traceability Matrix (RTM) detailed in Section 16.

14. Test Metrics & Reporting

Key metrics collected during execution include: percentage of test cases executed, pass/fail ratio, defect severity distribution, and requirement coverage. Daily summaries are reviewed during standups leading to the Final Test Summary Report.

15. Detailed Test Cases Suite

This test suite fulfills Deliverable 4 (Mini-Project Part-1), providing 10 comprehensive test cases covering both functional and non-functional requirements with prerequisites, test steps, test data, and expected outcomes.

Test ID & Type	Req ID & Module	Test Description & Preconditions	Execution Steps & Test Data	Expected Result
TC-Auth-01 Functional	CSQP-F-001 AuthModule	College Email Registration User does not exist in DB.	1. Open registration page. 2. Enter name, college email (.edu), password. 3. Submit form. 4. Check verification link.	Account created in unverified state; verification link emailed with 24h expiration.
TC-Auth-02 Functional / Security	CSQP-F-002 CSQP-SR-003 AuthModule	User Login & JWT Issuance User registered & verified.	1. POST /api/auth/login with valid email & password. 2. Inspect HTTP response headers and body.	HTTP 200 returned with signed JWT access token (1h expiry) and HttpOnly refresh cookie.
TC-Auth-03 Security / Functional	CSQP-F-003 CSQP-SR-004 AuthModule	API Role-Based Access Control Logged in as Student.	1. Send PATCH request to /api/questions/:id/status using Student JWT. 2. Check HTTP status code.	HTTP 403 Forbidden returned; question status remains unchanged in database.
TC-Sec-05 Security / NFR	CSQP-SR-005 AuthModule	Account Lockout After 5 Failed Logins Valid user account exists.	1. Submit 5 incorrect passwords sequentially. 2. Attempt 6th login with correct password within 15 minutes.	6th attempt rejected with account locked message; security audit log records lockout event.
TC-Mat-01 Functional	CSQP-F-010 MaterialModule	Study Material Upload Validation Authenticated Student.	1. Upload 30MB PDF file. 2. Upload 10MB PDF tagged with subject='CS101', topic='Trees'.	30MB upload rejected with size error; 10MB upload accepted and stored with topic tags.

15. Detailed Test Cases Suite (Continued)

Test ID & Type	Req ID & Module	Test Description & Preconditions	Execution Steps & Test Data	Expected Result
TC-Mat-04 Functional	CSQP-F-013 MaterialModule	Material Version History Retention Existing material version 1 exists.	1. Upload revised file for existing material. 2. Query versions array via API. 3. Download version 1.	Current version increments to 2; version 1 remains accessible and downloadable in history.
TC-QB-04 Functional	CSQP-F-020 CSQP-F-023 QuestionBank	Question Peer Submission & Approval Student and Instructor accounts.	1. Student posts new MCQ question. 2. Check quiz generation pool. 3. Instructor logs in and approves.	Question saved with status='pending' and excluded from quizzes until approved by instructor.
TC-Att-01 Functional	CSQP-F-030 QuizModule	Countdown Timer & Auto-Submit Student starts timed 10-minute quiz.	1. Start quiz session. 2. Answer 3 of 5 questions. 3. Let countdown reach 00:00 without clicking submit.	Attempt auto-submits at 00:00; answered questions are saved and graded accurately.
TC-Att-02 Functional	CSQP-F-031 GradingModule	Objective Auto-Grading & Review Flag Quiz with 4 MCQs and 1 Short Answer.	1. Submit completed responses. 2. Measure response time. 3. Check score breakdown.	Objective score appears in < 2 seconds; short-answer item is marked 'Pending Review'.
TC-Perf-01 Performance / NFR	CSQP-NF-001 All Modules	System Response Time Benchmark Staging environment under normal load.	1. Run k6 load test simulating 500 concurrent requests across login, search, and quiz submit.	90th percentile (p90) response time completes within 2 seconds without HTTP 500 errors.

16. Requirements Traceability Summary

Requirement ID	Requirement Category	Mapped Test Case(s)	Status
CSQP-F-001, 002, 003, 004	Authentication & Access Control	TC-Auth-01, TC-Auth-02, TC-Auth-03	Planned
CSQP-SR-001 to 005	Security & Account Lockout	TC-Auth-02, TC-Auth-03, TC-Sec-05	Planned
CSQP-F-010, 011, 012, 013	Study Material Management	TC-Mat-01, TC-Mat-04	Planned
CSQP-F-020, 021, 023, 024	Question Bank & Quiz Creation	TC-QB-04	Planned
CSQP-F-030, 031, 032, 033	Quiz Attempt, Anti-Cheat & Grading	TC-Att-01, TC-Att-02	Planned
CSQP-NF-001 to 005	Performance & Accessibility	TC-Perf-01	Planned